Critical Infrastructure Security (IY 5612/3612/4612) Syllabus for Academic Year 2025–2026

S. Wolthusen

January 5, 2026

1 Introduction

Cyber security can be considered as a specialised area of information security in that it focuses on an understanding of information systems linked with physical systems as is reflected by the etymology of the *cyber* prefix in κυβερνήτης who, fittingly, was not only the title of the helmsman but also the officer responsible for safety on a trireme. This is a characteristic shared with nearly all so-called *critical infrastructures*.

The objective of this module is hence to develop an understanding of advanced adversaries and threats to information systems, and particularly the interaction of information with physical systems at scales ranging from embedded and industrial control systems with particular emphasis on **critical infrastructures** at national and supra-national scales. Events in the past decade and more have perhaps confirmed that this is no longer an entirely hypothetical endeavour.

To this end, models of adversaries and interactions with adversaries will be studied together with approaches for modelling dependencies and interdependencies in networked systems including cyber-physical interactions. Based on this, advanced and persistent threats are studied, also to networks, embedded and control systems. The threat actors considered are demonstrably capable of also patiently using indirect approaches such as *supply chain attacks* which represent a challenge to those responsible for critical systems.

The module also aims to impart an understanding of assurance mechanisms for both the development and operation of security-sensitive and other high-integrity systems and will therefore provide a critical overview of relevant standards and evaluation criteria as well as of issues arising in the development of high assurance systems.

2 Learning Outcomes

Having completed the course, students should have

- gained an overview of core concepts of cyber security problems, and particularly their relation to critical (information) infrastructures and national cyber security strategies to defend such critical assets
- developed an understanding of dependencies and interdependencies in particularly larger-scale networks employing different modelling techniques
- the ability to analyse cyber-physical systems, particularly involving control systems and critical infrastructures, and to analyse attacks on such systems
- the ability to demonstrate critical awareness of advanced threats and attacks, and of methods for modelling and analysing such threats and attacks
- demonstrate a conceptual grasp of problems pertaining to specifying, measuring, and validating assurance including through formal assurance mechanisms
- the ability to evaluate problems and approaches related to information sharing, including security-related, legal, and ethical considerations
- a critical awareness of problems related to conflicts involving attacks on information systems including issues surrounding attribution and asymmetric conflicts

3 Formal Aspects

3.1 Faculty Contact Information

Stephen Wolthusen (main point of contact)

Office:	Bedford 2-31
Telephone:	01784 443 270
Email:	stephen.wolthusen@rhul.ac.uk

3.2 Prerequisites

The module benefits from a sound understanding of computer and network security. It is therefore desirable to have a general understanding of computer security concepts; in addition or in case that further material is required. Examples of background reading material at the advanced undergraduate or beginning postgraduate levels on on modern operating systems include the books by Tanenbaum [80] or Silberschatz *et al*. [72]. Similarly, it is also recommended to have read the module IY5511 (*Network Security*); in this case background reading on network security is e.g.

provided by the books of Stallings [74] or [75], while further general networking background can e.g. be found in the books of Comer [13].

3.3 Course Requirements and Objectives

Students will be expected to have a sound understanding of information security concepts as well as a solid grasp of core concepts in computer science and selected aspects of discrete mathematics mainly required for modelling as well as some understanding of elementary probability theory.

On completion of the module, students will have gained insight into selected aspects of cyber security beginning with high-level concepts such as national cyber security policies and their interactions with technical implementations. As the subjects of cyber security particularly at high levels will often exhibit dependencies, students will have gained exposure and the ability to analyse dependencies, and interdependencies using different modelling approaches varying in their level of detail including for critical infrastructures.

On completion of the module, students will further have achieved critical awareness of the cyber security issues found in cyber-physical systems and techniques employed in their security analysis including for sophisticated threats and attacks as well as incident response and recovery approaches. Students will also have gained a systematic understanding of assurance models and approaches for certification and accreditation.

3.4 Assessment and Academic Integrity

3.4.1 Assessment

The outcome of the course will be based on two assessment elements. An examination during the assessment period contributes 60% to the final outcome; the remaining assessment is formed by an essay (term paper) contributing 40% to the outcome[1].

3.4.2 Term Paper

The term paper is an essay on a topic that is within the scope of the module, but should delve into an area either covered by the lectures in some more depth, or should study an adjacent area; the topics identified in this syllabus documents can act as a guide.

If the chosen topic does not match with these requirements, it will be considered an invalid submission. Examples of possible term papers might include the study of a particular attack campaign including both technical

[1] Some candidates may be registered under old regulations (the IY5612O module code, only in this case is a separate piece of coursework required, this will take the form of a quiz.

and structural insights, or a review of methods for modelling dependencies in software, systems, or critical infrastructure not covered in the lecture notes. Where possible, primary literature (i.e. peer-reviewed research articles should be drawn upon).

Term papers will usually take the form of a *literature survey*, but should be based on one or two *research questions* to guide the selection of articles and book chapters used to construct the review and the formulation of the essay structure.

The length of the essay will vary depending on the module code under which the module is taken. For IY3612, the length should be 2000 words, for IY4612 the length should be 3000 words, and in case of IY5612 it should be 4000 words. The submission deadline is **26 March 2026**.[2]

As with any academic work, it is expected to be structured in support of its argument with clear and concise writing one of the key objectives. All arguments need to be supported, whether by direct evidence or proof, or more commonly by reference to the literature.

When relying on supporting evidence from the literature, every argument or claim should be supported by a clear, separate reference. The apparatus containing citations must follow a consistent format; whilst not required, use of the ACM citation style is recommended. When using LaTeX[3] for writing the term paper, this is supported by a dedicated BibTeX style.

3.4.3 Academic Integrity

Penalties will be imposed for academic dishonesty, with automatic referrals to appropriate college bodies possible depending on severity[4]. Academic dishonesty is defined as any action or practice that provides the potential for an unfair advantage to one individual or one group. This is in no small part to ensure that legitimate efforts are not invalidated.

Academic misconduct includes the misrepresentation of facts, the fabrication or manipulation of data or results, representing another's work or knowledge as one's own, disrupting or destroying the work of others, or abetting anyone who engages in such practices. Academic dishonesty is not absolute because the expectations for collaboration vary. However, unless given specific permission, any and all results submitted must be the result of individual effort, performed without the help of other individuals or outside sources, including the use of any automated text generation (large language models or similar) tools. Relying on tools for assistance with grammatical form or re-writing material does not absolve the candidate from responsibility to acknowledge the use of such methods or the

[2] Submissions are to be made electronically via Moodle with care to be taken to choose the appropriate submission box for the module code sat.

[3] Available for virtually all operating systems, see the LaTeX Project.

[4] See Royal Holloway's Regulations on Academic Misconduct

responsibility for the correctness of the content.

If a question arises about the type of external materials that may be used or the amount of collaboration that is permitted for a given task, each individual involved is responsible for verifying the rules with the lecturer or teaching assistant before engaging in collaborative activities, using external materials, or accepting help from others.

3.5 Schedule

Lectures will be given in the *second term*, beginning the week of January19, 2026 on *Monday*. Lectures will take place from 9am until 12 noon in the *Bourne Lecture Theatre 1*. To facilitate time-keeping, attendees should be in the lecture theatre well before the starting time to complete any necessary preparations and seating arrangements.

4 Required Reading

There is no single textbook on which the module is based; please refer to the notes for each unit. Several initial units make reference to chapters from [54]; note that these are available electronically through the college library subscription.

5 Lecture Schedule

The following sections contain both descriptions of the contents of each individual lecture together with required and recommended reading materials.

Lecture materials, exercise sheets, and a *mock exam* are posted on Moodle together with information on the blocks covered in each week's lecture. This mechanism also provides timely information on any changes to the schedule and should be reviewed regularly for any significant changes.

5.1 Unit 1: Introduction

Objectives

- Introduction to the concepts of cyber security and critical infrastructures
- Introduction to critical infrastructures, critical information infrastructures,
  and their relation to cyber security
- Overview of national cyber security strategies

Topics

Block 0 Introduction

- A sophisticated supply-chain attack: Sunburst
- Re-purposing ransomware for conflict: Colonial Pipeline ransomware, and criminal organisations in aid of the Russian attack on Ukraine
- Critical (information) infrastructure is also physical

Block 1 Administrative Matters

- Overview of the Critical Infrastructure Security module and topics covered
- A review of coursework and types of feedback provided

Block 2 Introduction to Cyber Security and Critical Infrastructures

- Defining Critical Infrastructures (CI) and
  Critical Information (CII) Infrastructures
- The relation between Cyber Security and CI/CII
- Threat actors and their classification

Block 3 National Cyber Security Strategies

- The relation between overall national security strategy and Cyber Security strategies
- The UK Cyber Security strategy
- Selected national Cyber Security strategies

Required and recommended reading The chapter by Dunn Cavelty and Suter provides an overview of the embedding of cyber security in the overall national security stratgy context and the relationship to critical infrastructures [19], while Luiijf provides a survey of cyber threats [56].

Strategy documents tend not to keep up well with developments, and it is at least as instructive to see what was considered vital just a few years earlier as it is to see current strategy.

The United Kingdom's overarching national security strategy in its publishable form is outlined in [32], while the UK Cyber Security strategy is found in [34, 35], building on [33]; the previous revisions and updates sketching out developments are found in [29, 31, 30].

The U.S. cyber security policy stance has shifted considerably in 2021 with Executive Order 14028 ("*Improving the Nation's Cybersecurity*") [91] with renewed emphasis not only on governmental and critical national infrastructure cyber security, but explicitly targeting the software supply chain security. This was also referenced in the most recent national cyber security strategy [83]. A more dated but comprehensive general U.S. policy perspective can be found in the 2009 review document [86], whilst the U.S. Department of Defense's published perspective particularly on operational considerations in the cyber domain was published in 2011 [84] with a more recent overview document on U.S. DoD strategy dating from 2015 [85]; strategies for protecting critical infrastructures were only published in draft form. Whilst the European Union does not have defence matters within its remit, several policy areas including related to national and European critical infrastructures such as payment and clearing systems, energy, and commerce are immediately affected; hence the Commission published a European Cyber Security strategy augmenting the respective national strategies is found in [36].

A more comprehensive survey of national cyber security strategies can be found in the NATO framework document [44]; related to this is a guide released by the ITU [93]; for a collection of national strategy documents see `https://ccdcoe.org/library/publications/national-cyber-security-strategy-guidelines/` with relevant NATO documents available in the NATO library `https://natolibguides.info/cyberdefence/documents`.

Some nation states have engaged in cyber operations for close to three decades; despite this the understanding of operations and their limitations is still very much evolving, including not only related to their effectiveness [27], but also where operational and legal constraints need to be considered [73]. Earlier perspectives were driven by the expectation of high-intensity interaction among nation-states, whereas more recently the emphasis has been on lower-level interactions [57] and the use of deniable assets [68].

The specific issue of subsea cabling infrastructure has recently received a significant amount of attention with events in the Baltic sea and is also described in e.g. in a report by ENISA [21]. A more global perspective and background can be found in [43].

5.2 Unit 2: Complex Attacks

Objectives

- Introduce complex attacks
- Understanding the cyber attack lifecycle and defences
- Case studies of cyber attack campaigns

Topics

Block 4 Complex Attacks

- The Cyber Kill Chain and MITRE Att&ck Framework
- Advanced persistent threat campaigns and selected countermeasures
- Deception defences

Block 5 APT Case Studies

- Early examples: Solar Sunrise, Moonlight Maze, and Titan Rain
- Aurora, APT1, and other "professional" groups
- Case study: The Regin toolkit
- State and non-state actors: Sofacy/Fancy Bear (APT28)
- Mobile Threats: Regin and Triangulation

Required and recommended reading For an early example of an espionage campaign conducted in computer networks see Stoll's entertaining account [76].

The concept of cyber kill chains was introduced in [40], while the book [41] offers some recent academic perspectives on key problems including attribution, deterrence, adversary profiling, and human factors; attribution is also studied in the article by Lindsay [52]. An unclassified view of cyber operations is provided by [87] while a discussion of related policy issues can be found in [81]; a NATO perspective on offensive operations is contained in [50] with a discussion on models of offensive operations in [25].

The more recent MITRE ATT&CK framework [78] offers a more narrative mechanism for describing adversary behaviour in the form of a taxonomy that had later been extended also to pre-attack behaviour [14]. This more extensive taxonomy is suitable for describing attacks, although for analysis it presents challenges in mapping activities [88]; it has since been extended inter alia to industrial control system attacks since the basic ATT&CK framework is somewhat domain-specific; this can also be used to construct adversary behaviour [71].

Some theoretical underpinnings on cyber attack can also be found in the article by Michael [59]; the Mandiant (purchased by FireEye in 2013 and de-merged again in 2021) report on the APT1 group and its activities

as well open source as attempts to identify these can be found in [22], while the Kaspersky analysis of the Regin platform is found in [42]. The APT28 group that seems to have played a certain role recently [90] is described in the FireEye report [23], while stealthy mechanisms using Twitter employed by the APT29 group as a vector are discussed in [24]. FireEye was, ironically, also one of the targets of the SolarWinds supply chain attacks in 2020–21 [89]. Mobile devices also figured in the 2023 Triangulation Attack [49], where the source is un-identified as of December 2023, but involving deep insights into hidden features of the Apple systems-on-a chip as well as a number of novel exploits.

Almost all campaigns highlighted in this unit are effectively intelligence operations; there is still an ongoing debate as to the discrepancy between the expected level of intensity of operations in the Russian attack against Ukraine and the observed impact as well as sophistication levels observed [48, 51]. However, some recent campaigns such as the "Salt Typhoon" activities in 2020–24 appear to be more in the nature of *pre-positioning*.

5.3 Unit 3: Networks and Dependencies

Objectives

- Develop systematic understanding of basic concepts in network modelling
- Conceptual understanding of network robustness and the effects of failures
  in networks for selected network models
- Synthesis of network robustness concepts developed for the case of the global
  Internet structure

Topics

Block 6 Models of Large-Scale Networks

- Elements and concepts in graph theory
- Methods for characterising graphs
- Basic network models: Random graphs

Block 7 Robustness of Networks

- Further properties and metrics for characterising graphs
- Matching existing physical and logical networks to the Erdős-Rényi random graph
- The Watts-Strogatz Small-World model
- Scale-free graphs and the Barabási-Albert graph model

Required and recommended reading A brief survey of network models is provided in the book chapter [79]. Whilst this is sufficient for the purposes of this lecture, a far more comprehensive and leisurely survey of networks and network science covering a number of ancillary areas is provided by [61] with chapters 7 and 8 of particular interest, while chapter 16 gives further insights into network resilience.

Although this module requires mainly a conceptual grasp of graph theory, the book by van Steen (on which the exposition of graph theoretic concepts is partly based) may provide valuable further reading at the advanced undergraduate level [92]. For further reading as well as a somewhat more rigorous but still accessible presentation, many alternative texts exist including [17].

5.4 Unit 4: Network and Internet Robustness

Objectives

- Understanding Internet Robustness to Random Faults and Attacks
- Robustness and attacks against the Internet infrastructure
- Understanding economic models of critical infrastructure dependencies

Topics

Block 8 Internet Robustness

- Modelling the global Internet structure
- An introduction to network robustness

Block 9 Robustness and attacks against the Internet Infrastructure

- Robustness of BGP routing
- Robustness of the DNS infrastructure
- Large-scale distributed denial of service attacks

Block 10 Economic Models of Critical Infrastructures

- Introduction to modelling approaches for dependencies and interdependencies
- Input-Output economic models
- Case Study: Centrality Measures for the Analysis of Financial Networks

Required and recommended reading Willinger and Roughan give a critical overview of Internet topology mapping research and associated problems [95]; at different abstraction levels this will yield quite variable insights and is subject to intensive research into methods for minimising any such errors. The only somewhat specious observation about the trade-off between the fidelity and necessary abstraction in the lecture was made by Wiener and Rosenblueth [65].

A seminal, but also often criticised, contribution on network robustness that is worth reviewing is the original article by Albert, Jeong, and Barabási [3], while the exposition of robustness aspects in the latter part of the unit is derived from the article by Deng *et al*. [16].

No current textbook covers routing protocols and security to a meaningful extent, so the primary RFC 4271 for BGP [64] may be reviewed. The survey papers [10, 39, 2] give a good insight into security problems and some approaches for protecting BGP from attacks and in part also misconfiguration. For in-depth but somewhat dated reading on routing protocols [26] or [38] may be more readable than the protocol specifications.

Economic and other qualitative as well as partially quantitative models of critical infrastructures are covered in the book chapter [79].

5.5 Unit 5: Critical Infrastructures and Interdependencies

Objectives

- Conceptual understanding of interdependency models at different abstraction levels
- Synthesis of dependency concepts and their effect on risk and vulnerability analysis in interdependent networks
- Understanding of power network concepts and their relation to critical infrastructures and control

Topics

Block 11 Critical Infrastructure Interdependency Models

- Graph-based models
- Flow problems and network robustness
- Agent-based models of interdependence
- Game-theoretical models

Block 12 Power Systems: A Case Study for Critical Infrastructures

- Power (and Smart) Grid Structrure
- Grid Requirements and Operation
- Communication and Control in the (Smart) Grid

The discussion of centrality measures for identifying systemically critical entities in the financial services sector is based on the article by Battiston *et al*. [7].

Cárdenas *et al*. discuss centrality and betweenness centrality results for a real communication network that would be considered carefully designed (yet still exhibits scale-free properties, as noted by the authors) [11]. For a study of betweenness measures for several electrical grids, consider the study by Hawick [28] detailing not only the immediate betweenness of a network, but also changes in betweenness on removal of some high-betweenness nodes as would be expected in an attack.

Schrijver describes the history of maximum flow problems arising from transportation networks and the study of their capacity and robustness [70].

The study of cascading failures has attracted considerable research interest, using a variety of formalisms and abstraction levels. The work by Buldyrev *et al*. [9] referenced in the lecture notes provides an example of this line of inquiry.

The lecture can give only a very brief motivation for the types of problems making agent-based modelling attractive in the analysis and simulation of computer and infrastructure networks; instead, only a simple example is provided. The main outline of the currently-used Tor protocol

(an overlay network intended to provide privacy and anonymity) as described by Dingledine and Mathewson [18] is analysed in an example of an agent-based simulation for studying strategies of enhancing the robustness to failure and attack when changing the entry-points ("guards") of the overlay by Elahi *et al*. [20].

Similar to agent-based models, the study of game-theoretical models for analysing security controls and properties is an active research area which is only introduced briefly; the specific question of resource allocation for security controls is e.g. studied by An *et al*. [6].

A basic introduction to power systems can e.g. be found in the work by Weedy *et al*. [94]

5.6 Unit 6: Security of Cyber-Physical Systems

Objectives

- Develop an understanding of key concepts in control systems theory and control systems architecture
- Synthesis of security challenges arising from cyber-physical interactions

Topics

Block 13 Introduction to Cyber-Physical Systems and Control Systems

- Control Systems Fundamentals
- Interaction Between Computation and Physical Environment
- Modelling, Specification, and Analysis of Cyber-Physical Systems

Block 14 Control Systems Architecture

- Basic Control Systems
- Components of Industrial Control Systems
- Real-Time Systems

Block 15 Attacks on Cyber-Physical Systems and Hybrid Attacks

- Attacks on Sensors and Actuators
- De-Synchronisation Attacks
- Altering Physical Parameters

Required and recommended reading The unit touches only on the very fundamentals of control systems theory and aims to be broadly qualitative, so if a deeper appreciation of problems and concepts is sought, this will require further self-study.

A very gentle introduction is e.g. the volume by Albertos and Mareels [4]; there exist a number of books on control systems theory where one approach may be more suitable to personal taste and background than another; one relatively comprehensive text at the advanced undergraduate level is the volume by Ogata [62]. While the title by Polderman and Willems is slightly older and currently out of print, it can be found online [63].

Chapter 2 of [77] gives a basic overview of the general control systems architecture. The bulk of [77] is concerned with control systems attacks; this will largely be covered in Unit 6 (see section 5.6), but for the purposes of this unit a more high-level perspective suffices.

A more in-depth view of attack modes on process control systems can e.g. be found in the article by Huang *et al*. [37].

5.7 Unit 7: Control Systems Security

Objectives

- Overview of basic concepts in control system implementations
- Conceptual understanding of state estimation problems
- Overview of security issues in control systems implementations

Topics

Block 16 SCADA and DCS Systems

- Introduction to Supervisory Control and Data Acquisition (SCADA) architectures
- Distributed Control Systems
- Control System Protocols

Block 17 State Estimation

- Linear State Estimators
- Non-Linear State Estimator
- Attacks on State Estimators

Block 18 Security of Control System Components

- Threats to Availability and Integrity
- Intrusion Detection in Control Systems

Required and recommended reading The NIST Special Publication 800-82 provides a good overview of current issues and challenges in control systems security [77].

A more network-oriented perspective is provided by Klein [45], with chapter 4 in particular dealing with the network protocols in industrial control systems. The UK CPNI also provides guidance for different aspects of control systems security [15].

A more in-depth view on some protocol vulnerabilities is e.g. provided by Rrushi in [67], while a discussion of cyber vulnerabilities in current (not necessarily "smart") power networks is provided by Bompard *et al*. in [8].

A set of examples of attacks on state estimation in power networks is found in the article by Liu *et al*. [53] (see [1] for a detailed monograph on power system state estimation).

A very brief outline of more practical attacks and techniques used in the analysis of control systems components is found in the white paper by Luallen [55], while some selected attacks on GPS systems are described in the paper by Tippenhauer *et al*. [82].

5.8 Unit 8: Attack Modelling Techniques

Objectives

- Survey of static risk and attack models
- Overview of selected dynamic attack and adversary models
- Introduction to the problem of concurrency in modelling adversary behaviour

Topics

Block 19 Static Risk and Attack Models

- Basic Risk Metrics
- Attack Trees

Block 20 Dynamic Attack and Adversary Models

- Attack-Defence Trees
- Game-Theoretic Models

Block 21 Concurrency in Adversary Models

- The problem of concurrency in attacker-defender interaction
- Petri nets for adversary modelling

Required and recommended reading A number of methods for analysing attacks based on approaches such as Fault Tree Analysis and Failure Mode and Effects Analysis have been described in the literature, partly intended for documenting analytical steps as in the case of [60], but also earlier e.g. in Amoroso's Threat Trees [5]. The term *Attack Tree* was then popularised by Schneier [69] and formulated rigorously by Mauw and Oostdijk [58].

Given the limitations of Attack Trees in becoming somewhat large and unwieldy for more complex scenarios and also the need to incorporate responses, a number of approaches have been put forward, including exensions of the earlier work by Mauw and Oostdijk by Kordy *et al*. [46] in the form of *Attack-Defence Trees*. This formalism can be extended further and has also been shown to be equivalent to a game-theoretical formulation.

A related approach which also includes a distinction between detection and mitigation events in the modelling of attacks was proposed by Roy *et al*. in the form of Attack Countermeasure Trees (ACT) [66]; as in the case of Attack Trees this allows both qualitative and quantitative study of attacks, including the cost effectiveness of detection and mitigation strategies where quantitative studies are used.

The article by Korzhyk *et al*. provides a summary of security games [47] and also contains further references to a number of existing models where different variants of security games are studied and deployed.

The article by Chen *et al*. gives a brief overview of different Petri net based concurrency formalisms for capturing concurrency in adversary behaviour, and also provides a case study in the cyber-physical interaction of the smart grid environment [12].

Reading List

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